

# Prashant Krishnan Vaidyanathan

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## Education

### University of California San Diego

San Diego, CA

M.S. in Computer Science and Engineering | GPA: 3.97

Sept 2021 - June 2023

- **Courses:** Algorithms, ML Data Systems, Advanced NLP, Recommender Systems, Programming Languages, Operating Systems, Adv. Text Mining
- **Graduate Student Researcher, Prof. Jingbo Shang:** Developed a robust framework for few-shot entity recognition for document images using multimodal BERT-like large language models and graph neural networks improving SOTA baselines such as LayoutLMv3. [\[Paper\]](#) [\[Code\]](#)

### Ramaiah Institute of Technology

Bangalore, India

B.E. in Computer Science and Engineering | GPA: 9.4/10

Aug 2015 - June 2019

## Work Experience

### Apple Inc.

Cupertino, CA

Machine Learning Data Scientist, Semantics (Apple Ads)

Jan 2026 - Present

- Developing an end-to-end LLM pipeline for semantic topic extraction, normalization, and evaluation of App Store apps at scale.
- Designed asymmetric loss functions for time series forecasting, improving inventory alignment with business goals
- Building an internal agentic LLM assistant for natural language querying, code generation, visualization and automated reporting

### Willot Inc.

San Diego, CA

Senior Data Scientist (Machine Learning Engineer)

June 2024 - Dec 2025

- Designed and implemented ETL pipelines and integrated custom XGBoost models, optimizing workflows for the largest retailers in the US
- Engineered an automated data labeling tool using weak supervision using Snorkel, increasing data by 2x & reducing manual costs by 5x.
- Spearheaded multiple releases, scaling deployment by 50x while collaborating with cross-functional teams to ensure performance & stability
- Executed the migration of ML pipelines to Databricks Unity Catalog, restructuring code, configuring access controls enhancing governance
- Optimized infrastructure by modifying Terraform scripts for ML deployments & transitioning to job clusters, reducing cloud costs by 35%

Data Scientist

Aug 2023 - May 2024

- Enhanced feature engineering strategy enabling near real-time predictions resulting in an improved model with latency reduced by 30%
- Implemented MLFlow for training, tracking, & logging ML experiments, including hyperparameter tuning, resulting in reproducible workflows.
- Explored use-cases of time series based large language models (LLMs) for ambient IoT data for classification and forecasting
- Refactored and modularized data labelling, preprocessing and analysis reducing processing time by 60% ensuring a streamlined workflow

### Apple Inc.

Austin, TX

Data Scientist Intern, Ad Platforms Data Insights

June 2022 - Sep 2022

- Created a scalable end-to-end pipeline for EDA, feature engineering and modeling for the top apps contributing to over ~ \$4B annually

### Indian Institute of Science

Bangalore, India

Research Engineer, LEAP Lab (Prof. Sriram Ganapathy)

Feb 2019 - June 2021

- Developed an end-to-end siamese neural network for backend modeling in speaker verification (~ 25% improvement) [\[Code\]](#)
- Built a diagnostic tool for COVID-19 using ML techniques on respiratory sounds with results matching rapid tests [\[Website\]](#) [\[Dataset\]](#)
- Led the collection of a multilingual multi-accent speaker profiling dataset with 300+ speakers and 6+ languages [\[Paper\]](#) [\[Dataset\]](#)

## Selected Publications (10+ Publications, 500+ citations)

- **P. Krishnan**, Z. Wang, Y. Wang, J. Shang, 'Towards Few-shot Entity Recognition in Document Images: A Graph Neural Network Approach Robust to Image Manipulation', LREC-COLING, 2024 [\[Paper\]](#) [\[Code\]](#).
- S. Ramoji, , **P. Krishnan**, and S. Ganapathy, "PLDA inspired Siamese Networks for Speaker Verification", Journal of Computer Speech & Language , Vol. 76, Nov 2022 p.101383. [\[Paper\]](#) [\[Code\]](#)
- N. Sharma, **P. Krishnan**, R. Kumar, S. Ramoji, S. R. Chetupalli, R. Nirmala, P. K. Ghosh, S. Ganapathy, "Coswara: A Database of Breathing, Cough, and Voice Sounds for COVID-19 Diagnosis", Interspeech 2020, Beijing. [\[Paper\]](#)

## Projects

- **Document to Image Generation** - Leveraged text mining techniques & large language models (LLMs) to generate images of long text documents such as poems and articles using Stable Diffusion and evaluated using OpenAI's CLIP score [\[Code\]](#)
- **Local RAG - Chat with your database:** Developed a local Retrieval-Augmented Generation (RAG) app, using LLMS such as Llama-3 & Mistral for queries about your PDFs, allowing database updates and using Streamlit for an intuitive UI [\[Code\]](#)
- **PDF Question Answer Chatbot** - Developed an interactive terminal application using LangChain and OpenAI's GPT APIs to extract embeddings from PDF documents, enabling users to retrieve information through natural language queries [\[Code\]](#)

## Skills

### Programming

Python, C, C++, R, HTML, JavaScript, TypeScript, Node.js, SQL, Git, Shell, Haskell

### Tools/Frameworks

PyTorch, PySpark, Tensorflow, Git, Databricks, Docker, AWS, GCP, Hugging Face, scikit-learn, Kubernetes